

Software Requirements Specification

Hakeem System

Jordan's National Electronic Health Record (EHR)

Scope, Functional Requirements, and Non-Functional Requirements

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the Hakeem System — Jordan's national Electronic Health Record (EHR) program. The document defines the scope of the system, its functional requirements, and the non-functional requirements that the system must satisfy. It is intended as a reference for stakeholders, including the Ministry of Health, Royal Medical Services, Electronic Health Solutions (EHS), participating healthcare facilities, system developers, and end users.

1.2 Background

The Hakeem program was launched in 2009 under the patronage of His Majesty King Abdullah II Bin Al Hussein, as the first national digital transformation initiative in Jordan's healthcare sector. It is managed by Electronic Health Solutions (EHS), a non-profit company designated by the Government of Jordan to implement and supervise the program nationwide.

Hakeem is built on a customized version of VistA (Veterans Health Information Systems and Technology Architecture), an open-source EHR platform originally developed by the U.S. Department of Veterans Affairs. VistA consists of more than 100 integrated software modules, which have been adapted by EHS to meet Jordan's regulatory, linguistic, and clinical needs.

The system has been deployed across Ministry of Health (MOH) hospitals, Royal Medical Services (RMS) facilities, the King Hussein Cancer Center (KHCC), and other healthcare institutions, with millions of patient records registered and tens of thousands of healthcare professionals trained on the platform.

1.3 Definitions and Acronyms

Term	Definition
EHR	Electronic Health Record
EHS	Electronic Health Solutions — the non-profit entity managing Hakeem
MOH	Ministry of Health of the Hashemite Kingdom of Jordan
RMS	Royal Medical Services
KHCC	King Hussein Cancer Center
VistA	Veterans Health Information Systems and Technology Architecture
CPRS	Computerized Patient Record System (the clinician-facing front end of VistA)
CPOE	Computerized Physician Order Entry
HL7	Health Level Seven — international standard for health data exchange
ELM	Electronic Library of Medicine

Term	Definition
HDA	Health Data Analytics program
BI	Business Intelligence
PDPL	Personal Data Protection Law No. (24) of 2023

1.4 Document Conventions

Functional requirements are identified using the prefix FR-x and non-functional requirements using NFR-x. Each requirement is presented in a table with a unique ID, description, and priority level (High, Medium, or Low).

2. System Scope

2.1 Overall Description

The Hakeem System is a nationwide Electronic Health Record platform that enables authorized healthcare providers across Jordan to electronically capture, store, manage, share, and analyze patient medical information. Patients are uniquely identified by their National ID number, which serves as the primary key linking all of their clinical records across participating healthcare facilities.

Hakeem provides a unified, longitudinal medical record that follows the patient between primary care centers, specialty clinics, and hospitals. Clinicians use the system to document encounters, prescribe medications, order laboratory and radiology investigations, view results, and make clinical decisions based on a complete view of the patient's history.

2.2 In-Scope

The Hakeem System covers the following capabilities and stakeholders:

- Nationwide electronic health record covering public healthcare facilities (MOH, RMS, KHCC) and participating private institutions.
- Comprehensive clinical documentation: medical and surgical history, physical examinations, procedural and surgical reports, vital signs, allergies, and progress notes.
- Order management for medications, laboratory tests, radiology investigations, dietary orders, and clinical procedures (CPOE).
- Pharmacy management, including outpatient and inpatient dispensing, drug interaction checks, and prescription tracking.
- Laboratory and radiology information systems integration, including result reporting and image viewing via VistA Imaging.
- Patient registration, admission/discharge/transfer (ADT), and appointment scheduling.
- Patient billing and stock management integration via HL7 messaging through the EHS HL7 Framework.
- Role-based access for physicians, nurses, pharmacists, medical technologists, administrators, and other clinical staff.
- Patient-facing access through the "My Hakeem" web and mobile service, enabling patients to view selected portions of their health record.
- Health Data Analytics (HDA) capabilities, including Business Intelligence and Artificial Intelligence applied to anonymized data for research and policy.
- Integration with the Electronic Library of Medicine (ELM) for clinical decision support and evidence-based medicine.
- Compliance with Jordan's Personal Data Protection Law No. (24) of 2023 and other applicable national legislation.

2.3 Out-of-Scope

- Health insurance claims adjudication for private insurance providers (handled by external systems).
- Financial accounting and human resource systems of healthcare facilities (integrated only at the data-exchange level).
- Direct medical device control or real-time life-support automation.
- Public health surveillance systems beyond the data feeds Hakeem provides to authorized authorities.
- Telemedicine consultation infrastructure (video, audio) unless explicitly contracted as a Hakeem module.

2.4 User Classes

User Class	Description
Physicians	Diagnose, document encounters, place orders, review results, and prescribe medications.
Nurses	Document vital signs, administer medications, record nursing notes, and execute care plans.
Pharmacists	Verify, dispense, and track medications; review drug interactions and patient allergies.
Medical Technologists	Process laboratory orders, record results, and validate findings.
Radiologists	Read imaging studies, document reports, and link images through VistA Imaging.
Administrative Staff	Manage registration, scheduling, admissions, discharges, and billing.
System Administrators	Manage users, roles, system configuration, integration, and audit logs.
Patients	Access selected personal health information through the "My Hakeem" portal.
Researchers / Policy Makers	Access anonymized aggregated data through the HDA platform.

3. Functional Requirements

The following requirements describe what the Hakeem System shall do. Requirements are grouped by functional area.

3.1 Patient Registration and Identification

ID	Requirement Description	Priority
FR-01	The system shall register every patient using the Jordanian National ID number as the unique identifier. For non-Jordanian patients, an alternative unique identifier (passport number) shall be supported.	High
FR-02	The system shall capture and maintain demographic data: full name (Arabic and English), date of birth, gender, nationality, contact details, address, and emergency contact.	High
FR-03	The system shall detect and prevent duplicate patient records by validating identifiers against the central registry.	High
FR-04	The system shall classify patient records as sensitive or non-sensitive and allow the patient to lock their data upon request.	High

3.2 Clinical Documentation

ID	Requirement Description	Priority
FR-05	The system shall allow authorized clinicians to record the complete medical and surgical history of patients, including chronic conditions and family history.	High
FR-06	The system shall record physical examination findings, vital signs, allergies, and immunization history.	High
FR-07	The system shall provide structured templates and free-text fields for progress notes, discharge summaries, and operative reports.	High
FR-08	The system shall maintain an auditable, longitudinal record that follows the patient across all participating facilities.	High
FR-09	The system shall support clinical coding using ICD-10 for diagnoses and standard nomenclatures for procedures.	Medium

3.3 Computerized Physician Order Entry (CPOE)

ID	Requirement Description	Priority
FR-10	The system shall allow physicians to electronically place orders for medications, laboratory tests, radiology investigations, diet, and clinical procedures.	High
FR-11	The system shall validate orders against patient allergies, current medications, and known drug-drug interactions, and shall display alerts to the prescriber.	High

ID	Requirement Description	Priority
FR-12	The system shall route orders automatically to the appropriate department (pharmacy, laboratory, radiology) for execution.	High
FR-13	The system shall provide order sets and clinical protocols configurable per facility or specialty.	Medium

3.4 Pharmacy Management

ID	Requirement Description	Priority
FR-14	The system shall manage outpatient and inpatient prescriptions from order entry through dispensing.	High
FR-15	The system shall maintain a national drug formulary and link prescriptions to the formulary.	High
FR-16	The system shall track medication administration, including time, dose, route, and the administering clinician.	High
FR-17	The system shall provide pharmacy inventory and stock management integrated via HL7 messaging.	Medium

3.5 Laboratory and Radiology

ID	Requirement Description	Priority
FR-18	The system shall accept laboratory test orders, route them to laboratory analyzers, and return results electronically.	High
FR-19	The system shall integrate with VistA Imaging to display radiology images and scanned documents linked to the patient record.	High
FR-20	The system shall allow radiologists and pathologists to issue, sign, and amend reports electronically.	High
FR-21	The system shall flag abnormal or critical results and notify the responsible clinician.	High

3.6 Admission, Discharge, and Transfer (ADT) and Scheduling

ID	Requirement Description	Priority
FR-22	The system shall manage patient admissions, bed assignments, transfers between wards, and discharges.	High
FR-23	The system shall support outpatient appointment scheduling for clinics, including resource and physician calendars.	High
FR-24	The system shall record discharge summaries and post-discharge follow-up plans.	Medium

3.7 Integration and Interoperability

ID	Requirement Description	Priority
FR-25	The system shall exchange clinical and administrative data with other systems using HL7 v2 messaging through the EHS HL7 Framework.	High
FR-26	The system shall integrate with the patient billing systems of participating facilities.	High
FR-27	The system shall integrate with hospital stock management systems for pharmacy and supplies.	Medium
FR-28	The system shall expose APIs that support standards-based interoperability with national health platforms.	Medium

3.8 Patient Portal (My Hakeem)

ID	Requirement Description	Priority
FR-29	The system shall provide a secure web and mobile patient portal ("My Hakeem") allowing patients to view selected portions of their electronic health record.	High
FR-30	The system shall authenticate portal users using a secure mechanism (e.g., National ID and verified contact information).	High
FR-31	The system shall allow patients to request access controls on sensitive information.	High

3.9 Analytics and Reporting (HDA)

ID	Requirement Description	Priority
FR-32	The system shall provide a Health Data Analytics platform that consumes anonymized data from Hakeem and produces dashboards, reports, and indicators.	High
FR-33	The system shall support epidemiological reporting, disease registries (e.g., Hemophilia, oncology), and the National Health Registry.	High
FR-34	The system shall apply Business Intelligence and AI techniques to support clinical research and decision-making.	Medium

3.10 Administration, Security, and Audit

ID	Requirement Description	Priority
FR-35	The system shall provide role-based access control with predefined roles for clinicians, administrators, pharmacists, and technologists.	High

ID	Requirement Description	Priority
FR-36	The system shall maintain a tamper-evident audit log of all data access and modifications, identifying the user, action, timestamp, and patient record involved.	High
FR-37	The system shall provide tools for system administrators to manage users, roles, facilities, drug formularies, and clinical templates.	High
FR-38	The system shall provide a training environment separate from production for Hakeem Academy and end-user training.	Medium

4. Non-Functional Requirements

These requirements describe how the system must behave in terms of quality attributes such as performance, security, reliability, and usability.

4.1 Performance

ID	Requirement Description	Priority
NFR-01	The system shall return a patient's summary record within 3 seconds when queried by National ID under normal load conditions.	High
NFR-02	The system shall support concurrent use by at least the full population of trained Hakeem users across MOH, RMS, and KHCC facilities without measurable degradation.	High
NFR-03	Standard clinical transactions (order entry, note saving, result retrieval) shall complete within 5 seconds at the 95th percentile.	High

4.2 Availability and Reliability

ID	Requirement Description	Priority
NFR-04	The core clinical components of Hakeem shall be available 24 hours a day, 7 days a week, with a target availability of at least 99.5% per year, excluding planned maintenance.	High
NFR-05	Planned maintenance windows shall be scheduled outside peak clinical hours and communicated to facilities in advance.	Medium
NFR-06	The system shall support a disaster recovery site with a Recovery Time Objective (RTO) of no more than 4 hours and a Recovery Point Objective (RPO) of no more than 15 minutes for clinical data.	High

4.3 Scalability

ID	Requirement Description	Priority
NFR-07	The system architecture shall scale horizontally to accommodate growth in patient population, healthcare facilities, and user base without redesign.	High
NFR-08	The system shall be able to store and retrieve clinical records for millions of patients over multi-decade timeframes.	High

4.4 Security

ID	Requirement Description	Priority
NFR-09	The system shall comply with Jordan's Personal Data Protection Law No. (24) of 2023, including lawful processing, consent management, breach notification, and data subject rights.	High
NFR-10	All communication between client applications, integration interfaces, and servers shall be encrypted using industry-standard transport security (TLS).	High
NFR-11	Sensitive personal data (including psychiatric, sexual health, and genetic data) shall be classified and protected by additional access controls.	High
NFR-12	The system shall require strong authentication for all users; passwords shall meet defined complexity rules and be renewed periodically.	High
NFR-13	The system shall log all access to patient records and shall make audit data available to authorized auditors and the Data Protection Officer.	High
NFR-14	Patients shall have the right to lock their records and to be notified of any data breach affecting their data, in accordance with PDPL Article 20.	High

4.5 Privacy and Compliance

ID	Requirement Description	Priority
NFR-15	The system shall obtain and record patient consent where required by law, in accordance with Articles 4 and 5 of the PDPL.	High
NFR-16	Data shall not be transferred outside the Kingdom of Jordan unless the receiving party provides a level of protection equivalent to that required by the PDPL, except in cases authorized by Article 15.	High
NFR-17	The Controller (EHS / participating facility) shall appoint a Data Protection Officer per Article 11 of the PDPL.	High

4.6 Usability

ID	Requirement Description	Priority
NFR-18	The user interface shall support both Arabic and English, with the ability for users to switch language at runtime.	High
NFR-19	The interface shall follow consistent clinical workflows so that staff trained on one Hakeem facility can use the system at any other Hakeem-enabled facility.	High
NFR-20	The system shall provide context-sensitive help and integration with the Electronic Library of Medicine (ELM) for clinical decision support.	Medium
NFR-21	Training programs (Hakeem Academy) shall be provided to ensure that all categories of users can use the system safely.	High

4.7 Interoperability and Standards

ID	Requirement Description	Priority
NFR-22	The system shall comply with HL7 v2 messaging standards for clinical and administrative data exchange.	High
NFR-23	The system shall support standard clinical terminologies (ICD-10 for diagnoses; LOINC and SNOMED CT as adopted by the MOH).	High
NFR-24	The system shall provide imaging support via DICOM through VistA Imaging.	High

4.8 Maintainability and Open-Source Governance

ID	Requirement Description	Priority
NFR-25	The system shall be based on the WorldVistA open-source platform and shall use the UJO7 namespace assigned to Jordan to avoid conflicts with the global VistA codebase.	High
NFR-26	Customizations developed by EHS shall be documented and, where appropriate, contributed back to the open-source VistA community under the GPL license.	Medium
NFR-27	The system shall be designed in modules so that individual components can be upgraded independently without service-wide downtime.	Medium

4.9 Backup and Data Retention

ID	Requirement Description	Priority
NFR-28	Clinical data shall be backed up on a regular schedule, with backup copies stored in geographically separate locations.	High
NFR-29	Data retention periods shall comply with Jordanian medical record retention regulations and Article 6(B) of the PDPL (which prohibits retention beyond the purpose of processing, unless other legislation requires it).	High

4.10 Portability and Deployment

ID	Requirement Description	Priority
NFR-30	The system shall be deployable in both centralized and distributed configurations across MOH, RMS, and KHCC infrastructure.	Medium
NFR-31	Client workstations shall be supported on standard wired and wireless network configurations using current operating systems used in Jordanian healthcare facilities.	Medium

5. Constraints and Assumptions

5.1 Constraints

- The system must operate within Jordan's legal and regulatory environment, including the Personal Data Protection Law No. (24) of 2023 and Ministry of Health regulations.
- The system is based on the VistA open-source codebase; major architectural changes must remain compatible with the global VistA community and the UJO7 namespace.
- Implementation depends on the availability of network infrastructure at each healthcare facility.
- Clinical workflows must be adapted to local medical practice and the Arabic language.

5.2 Assumptions

- All participating healthcare facilities will provide accurate patient demographic data through the National ID system.
- Healthcare professionals using the system have received training through Hakeem Academy or equivalent programs.
- Integration partners (laboratory analyzers, billing systems, imaging modalities) support HL7 v2 and DICOM standards.
- The Ministry of Health, RMS, KHCC, and EHS provide ongoing governance through the appropriate steering committees.

6. References

- Electronic Health Solutions (EHS) — Hakeem Program documentation: <https://ehs.com.jo/hakeem-program>
- Electronic Health Solutions (EHS) — VistA implementation in Jordan: <https://ehs.com.jo/hakeem-program/vista>
- My Hakeem patient portal: <https://my.hakeem.jo>
- Hashemite Kingdom of Jordan — Personal Data Protection Law No. (24) of 2023.
- U.S. Department of Veterans Affairs — VistA (Veterans Health Information Systems and Technology Architecture).
- WorldVistA open-source EHR platform.